



GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY

Department of Science and Technology

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ROBOFEST GUJARAT – 6.0



EKAGRATA: AI-POWERED ATHLETE

**DIGITAL TWIN AND
MOTION SHADOW FOR
VIRTUAL PERFORMANCE
OPTIMIZATION**

Ideation Stage / Concept, Robofest 6.0

Technical Details for Hackathon

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I. Type of Robot

- **TYPE OF ROBOT: Autonomous Sports Performance Monitoring and Optimization Robot**

A. Overview of the Proposed Robot

EKAGRATA is an AI-powered robotic system designed to assist athletes in improving performance through real-time motion tracking, sensor fusion, digital twin technology and virtual simulation. A camera and lightweight wearable IMU sensors capture the athlete's movement, while computer vision and motion analysis estimate posture, joint angles, orientation and movement dynamics.

The captured data is used to create a personalized athlete digital twin inside a virtual sports environment. A real-time Motion Shadow mirrors the athlete's movement, allowing actual and virtual movements to be compared. The system analyses fine-grained biomechanical parameters such as joint movement, timing, velocity, acceleration and trajectory.

EKAGRATA then performs controlled what-if simulations by modifying selected movement parameters and uses an AI-based optimization system to identify potentially improved movement configurations. The resulting visual and quantitative insights are provided to the athlete and coach for further training and validation. The architecture is designed to support multiple sports, with the initial prototype focused on one selected sport.

B. Key Capabilities

- **Real-Time Motion Tracking & Sensor Fusion:** Captures athlete movements using camera-based computer vision and wearable IMU sensors.
- **Athlete Digital Twin:** Creates an athlete-specific virtual model representing the individual's body configuration, movement patterns and performance parameters.
- **Real-Time Motion Shadow:** Mirrors the athlete's physical movement within the virtual environment, enabling direct visual comparison between actual and digital motion.
- **Biomechanical Performance Analysis:** Analyses key parameters such as joint angles, movement timing, velocity, acceleration, body alignment and movement trajectories.
- **Virtual What-If Simulation & AI Optimization:** Modifies selected movement parameters in the virtual environment and uses AI-based optimization to evaluate potentially improved configurations.
- **Continuous & Multi-Sport Performance Improvement:** Uses subsequent training data to update the digital twin and provides a scalable framework that can be adapted to different sports through sport-specific models.

C. Type:

Both — Primarily Simulation with Limited Hardware Integration

The project is predominantly simulation-based, with a small hardware sensing layer consisting of a camera and lightweight wearable IMU sensors for real-time athlete motion capture. The major processing, digital twin, Motion Shadow, virtual sports environment and AI-based optimization will be implemented in simulation.

D. List of Software's to be Used:

1. Blender:

Blender will be used to create the virtual athlete, digital twin and sports environment. It will also be used to visualize the athlete's Motion Shadow and compare the original movement with the simulated movement.

License: Free and open-source under the GNU GPL.

Our Contribution: We will create the athlete model, rig, virtual environment and movement visualizations required for the project.



Figure 1: Blender

2. Python

Python will be the main programming language for EKAGRATA. It will be used for processing sensor data, analyzing motion, connecting different modules and implementing the AI and optimization algorithms.

License: Python Software Foundation License.

Our Contribution: We will develop the main processing pipeline, data handling, analysis and integration logic.



Figure 2: Python

3. OpenCV

OpenCV will be used to process the video received from the camera. It will handle tasks such as frame processing, image preprocessing and extracting useful visual information from the athlete's movement.

License: Apache License 2.0.

Our Contribution: We will develop the vision pipeline and connect the camera data with the motion-analysis system.



Figure 3: Opencv

4. Media Pipe

Media Pipe will be used to detect the athlete's body landmarks from the camera feed. These landmarks will provide the positions needed to calculate joint angles and reconstruct the athlete's movement.

License: Apache License 2.0.

Our Contribution: We will integrate the pose-estimation output with our motion reconstruction and digital-twin pipeline.

5. PyTorch

PyTorch will be used wherever machine-learning models are required for movement analysis, performance assessment and optimization.

License: Open-source, BSD-style licensing.

Our Contribution: We will develop, train and integrate the required AI/ML models with the simulation and performance-analysis modules.



Figure 4: Pytorch

6. ROS 2

ROS 2 will provide the communication framework between the sensing system, processing modules, digital twin and virtual robotic system. It will help us keep the different parts of EKAGRATA connected in modular architecture.

License: Open-source; individual ROS 2 packages are distributed under their respective licenses.

Our Contribution: We will develop the required ROS 2 nodes, communication interfaces and data flow between the system components.

7. Gazebo

Gazebo will be used to simulate the virtual robotic athlete and its interaction with the sports environment. It will allow us to test movement, sensors and control algorithms without depending entirely on physical trials.

License: Apache License 2.0.

Our Contribution: We will create the simulated robot model, sensor setup and virtual environment required for testing.

8. NumPy

NumPy will be used for numerical calculations involved in motion analysis, including joint-angle calculations, matrix operations, trajectories and sensor-data processing.

License: Modified BSD License.

Our Contribution: We will implement the mathematical calculations needed for motion-analysis and optimization modules.

II. Robot Assembly Design

A. Mechanical CAD Overview

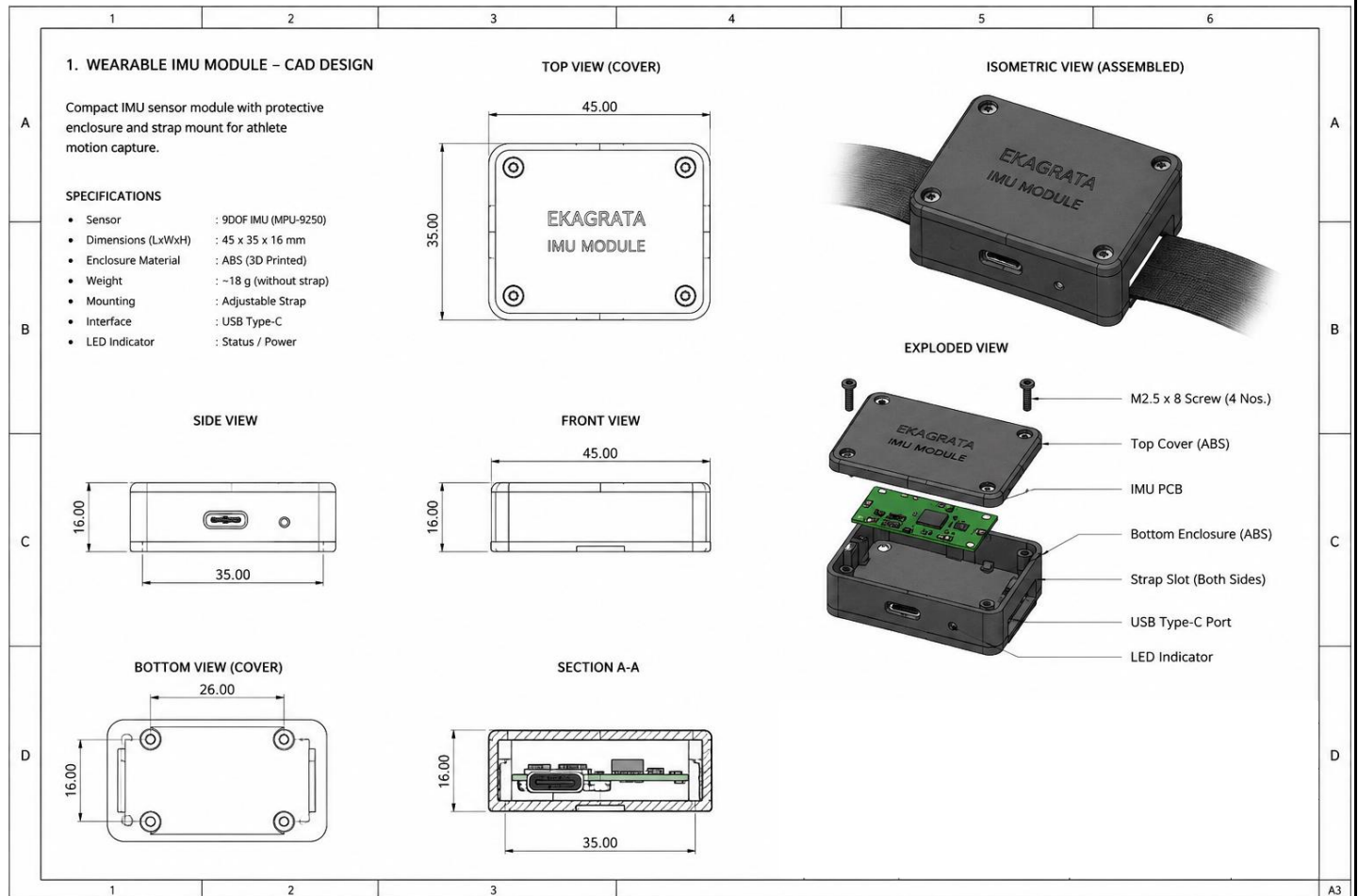


Figure 5: Wearable IMU Sensor Module – EKAGRATA

B. Calculations

D. 1 IMU Sampling Frequency

To capture rapid athletic movements, the sensor sampling frequency must be sufficiently higher than the highest frequency component of the motion.

Using the Nyquist criterion:

$$f_s \geq 2f_{\max}$$

For the proposed system, assuming the highest relevant motion component is approximately:

$$f_{\max} = 20\text{Hz}$$

The minimum theoretical sampling frequency is:

$$f_s \geq 2(20) = 40\text{Hz}$$

A **100 Hz sampling rate** is therefore proposed for the wearable IMU layer to provide additional temporal resolution for motion analysis.

$$f_s = 100\text{Hz}$$

This gives a sampling interval of:

$$\Delta t = \frac{1}{100} = 0.01\text{s}$$

Thus, the system obtains a new motion measurement every **10 ms**.

D.2 Joint Angle Calculation

The joint angle is calculated from three body landmarks. For example, for the elbow:

A = shoulder

B = elbow

C = wrist

The angle at the elbow is:

$$\theta = \cos^{-1} \left(\frac{(A - B) \cdot (C - B)}{|A - B||C - B|} \right)$$

For example, if:

$$A=(0.20, 1.50), B=(0.35, 1.25), C=(0.55, 1.20)$$

then the vectors are:

$$\overrightarrow{BA} = A - B = (-0.15, 0.25)$$

$$\overrightarrow{BC} = C - B = (0.20, -0.05)$$

Using the dot-product relation:

$$\theta = \cos^{-1} \left(\frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} \right)$$

This produces the instantaneous elbow angle used by the biomechanical analysis module.

D. 3

Motion Shadow Reconstruction Error

The accuracy of the digital twin can be evaluated by comparing the real athlete's detected landmark position with the corresponding Motion Shadow position.

For a point in 3D:

$$E_p = \sqrt{(x_r - x_s)^2 + (y_r - y_s)^2 + (z_r - z_s)^2}$$

Where:

x_r, y_r, z_r = measured athlete position

x_s, y_s, z_s = Motion Shadow position

For example, if:

$$P_r = (1.20, 0.85, 1.60)$$

and

$$P_s = (1.18, 0.88, 1.57)$$

then:

$$E_p = \sqrt{(1.20 - 1.18)^2 + (0.85 - 0.88)^2 + (1.60 - 1.57)^2}$$

$$E_p = \sqrt{0.0004 + 0.0009 + 0.0009}$$

$$E_p \approx 0.047 \text{ m}$$

So, the positional reconstruction error is approximately:

$$\boxed{4.7 \text{ cm}}$$

III. Components to be used

A. List of Structure components

Sr. No.	Name	Purpose	Image
1.	IMU Sensor	Protects and mounts the wearable IMU	
2.	Adjustable Sensor Strap	Secures IMU modules to selected body segments	
3.	Soft Padding	Improves comfort and reduces sensor vibration	

B. List of motion

Table 1: List of Structure components

components

Sr. No.	Name	Purpose
1.	Virtual Joint Actuators	Simulates movement of the virtual robotic athlete
2.	Virtual Joint Controllers	Controls joint position, velocity and trajectory in simulation

Table 2: List of motion components

C. List of Electronic components

Sr. No.	Components
1.	Raspberry pi 5
2.	ESP32-DevKitC
3.	BNO055 9-axis IMU
4.	C920s PRO HD Webcam
5.	USB Type-C Power Supply
6.	Jumper / Interface Cables

Table 3: List of electronic components

1. Raspberry pi 5 (Model B-8GB)

Raspberry Pi 5 will act as the primary edge-computing unit of EKAGRATA. It will receive and process motion data from the wearable sensor modules and camera, perform initial data handling, and communicate the processed information with the digital-twin and simulation environment. Its processing capability and support for high-speed connectivity make it suitable for real-time motion monitoring and integration of the sensing, AI and simulation modules.



Figure 6: Raspberry pi 5

Specifications:

Processor	Broadcom BCM2712, Quad-core 64-bit Arm Cortex-A76 @ 2.4 GHz
Power	5V 3A (USB Type-C / GPIO / PoE)
GPIO	40-pin header
Multimedia	4Kp60 HEVC decode, OpenGL ES 3.1, Vulkan support
Video / Audio	Dual micro-HDMI (4Kp60), CSI/DSI ports, stereo audio
Connectivity	USB, Ethernet, Wi-Fi, Bluetooth

Table 4: Specifications of Raspberry pi

2. ESP 32-DevKitC

The ESP32-DevKitC will be used as the local controller for EKAGRATA's wearable motion-sensing units. It will collect data from the connected IMU sensors and transmit the sensor readings to the Raspberry Pi 5 through wireless communication. The board provides the required GPIO, communication interfaces and processing capability for handling multiple sensor inputs, making it suitable for developing a compact and modular wearable sensing system. It will also support initial sensor-data processing and synchronization before the data is integrated into the athlete digital twin.



Figure 7: ESP32-DevKitC

Specifications:

Model	ESP32 Series SoC
Input Voltage	3.3 V logic
Processor	Dual-core Xtensa 32-bit LX6, up to 240 MHz
USB Interface	Micro-USB for power, programming and serial communication
Flash Memory	Typically 4 MB on standard DevKitC variants

Table 5: Specifications of ESP 32- DevkitC

3. BN0055 9-axis IMU

The **BN0055** from Bosch Sensortec will be used as a wearable motion sensor for EKAGRATA. It integrates a **triaxial accelerometer, triaxial gyroscope and triaxial geomagnetic sensor** with an onboard microcontroller and sensor-fusion software, allowing orientation and motion information to be obtained directly from the sensor. This makes it suitable for tracking the orientation and dynamic movement of selected athlete body segments such as the torso, arms and legs. The sensor supports fused orientation outputs at up to **100 Hz**, which is suitable for the proposed real-time motion-tracking system.

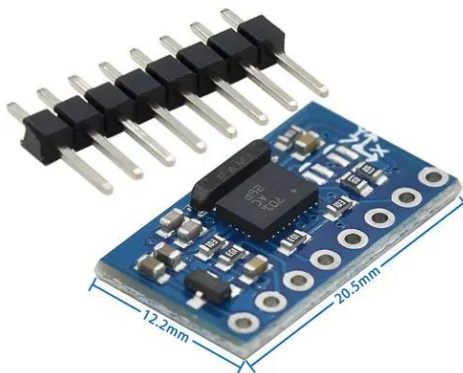


Figure 8: 3.BN0055 9-axis IMU

Specifications:

Sensor Type	9-axis absolute orientation sensor
Accelerometer	Triaxial, selectable $\pm 2\text{ g}$ / $\pm 4\text{ g}$ / $\pm 8\text{ g}$ / $\pm 16\text{ g}$
Gyroscope	Triaxial, 16-bit
Control Signal	PWM (1100 μs reverse, 1500 μs neutral, 1900 μs forward)
Output Data	Euler angles, quaternion, rotation vector, linear acceleration and gravity

Table 6: Specifications of BN005

4. C920s PRO HD Webcam

The Logitech C920s PRO HD Webcam will be used as the primary visual sensing device for EKAGRATA. It will capture the athlete's movements during training and provide video input for the computer-vision pipeline. The captured frames will be processed using OpenCV and MediaPipe to detect body landmarks, estimate pose and reconstruct movement parameters required for the athlete's digital twin and real-time Motion Shadow. Its autofocus capability and Full HD video output make it suitable for consistent athlete-motion capture during the proposed experiments.



Figure 9: C920s PRO HD Webcam

Specifications:

Image Sensor	Full HD glass lens with autofocus
Maximum Resolution	1920 × 1080 pixels
Video Format	H.264, MJPEG, YUY2
Connection	USB-A
Field of View	78° diagonal
Focus Type	Autofocus

Table 7: Specifications of C92

5. USB Type-C Power Supply

The USB Type-C Power Supply will provide a stable and regulated power source for the Raspberry Pi 5 and associated processing components of EKAGRATA. It will support reliable operation during continuous camera capture, sensor-data acquisition, motion processing and communication with the digital-twin system. The power supply will be selected according to the required voltage and current ratings of the connected hardware to ensure safe and stable operation during testing and simulation-assisted athlete monitoring.



Figure 10: Power supply

IV. Methodology of Making Robot

A. Overall System Methodology

EKAGRATA is developed as a **hybrid physical-digital robotic system** in which a limited physical sensing layer captures the athlete's movement and a virtual robotic system performs detailed analysis and simulation.

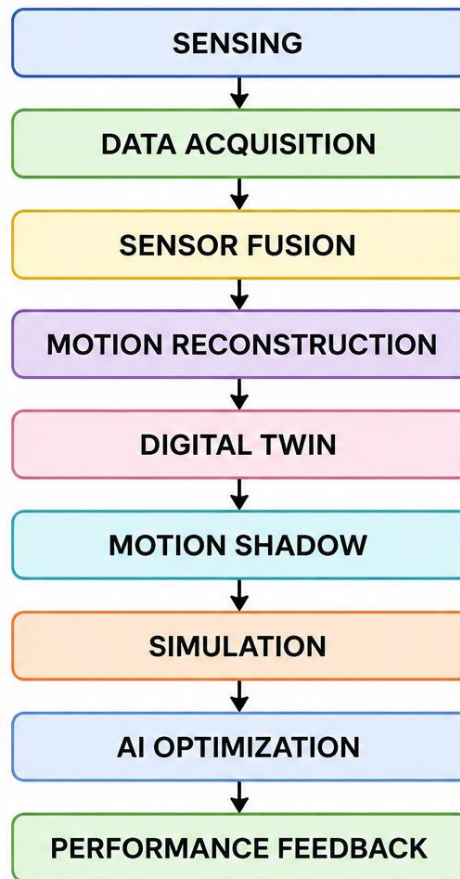


Figure 11: Overall Methodology of EKAGRATA

B. Physical Sensing Methodology

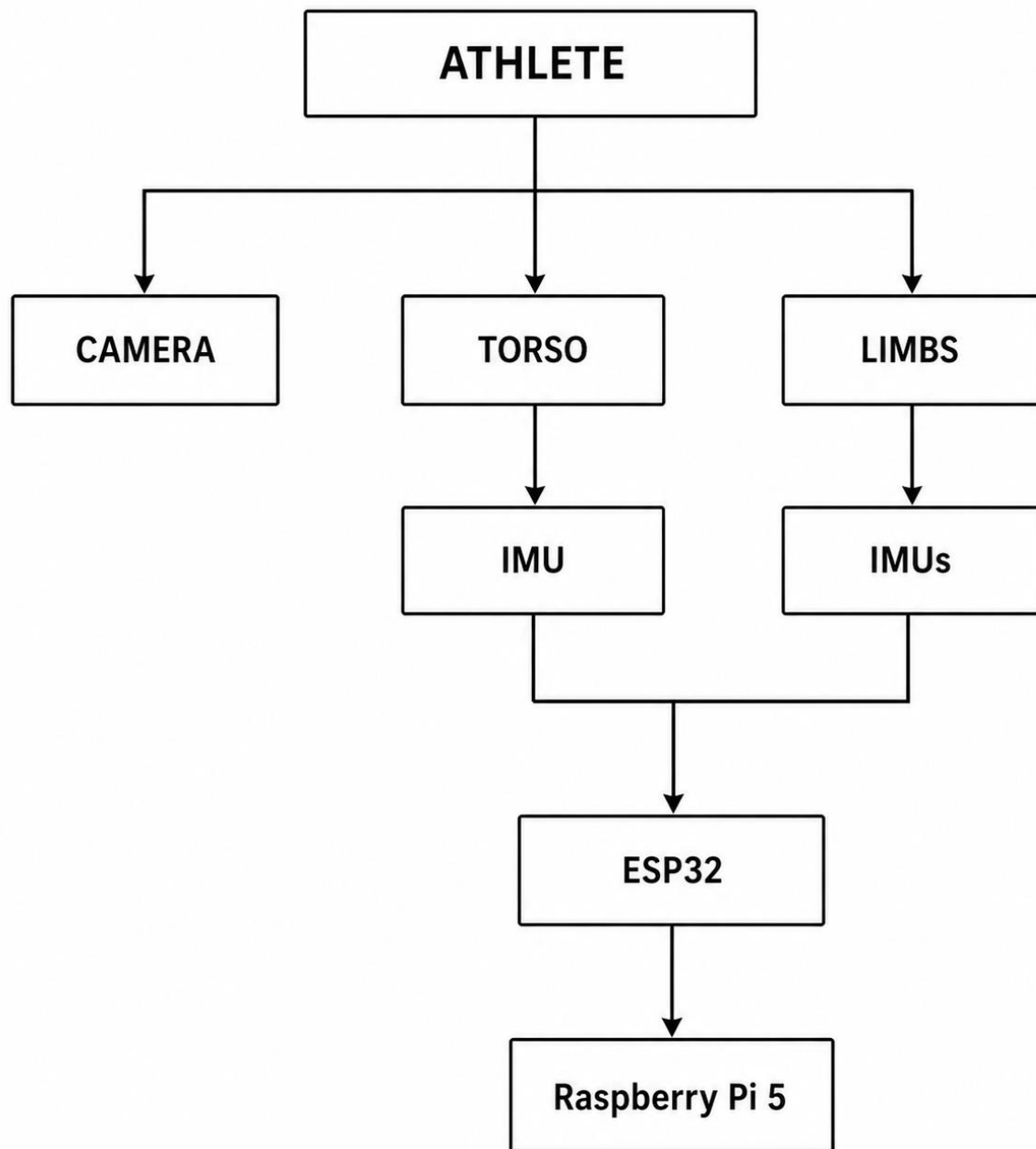


Figure 12: Physical Sensing Methodology

The physical sensing layer consists of a Logitech C920s webcam, wearable BNO055 IMU modules, ESP32-based sensor nodes and Raspberry Pi 5.

The camera continuously captures the athlete's movement, while the wearable IMUs measure orientation and motion of selected body segments.

C. Data Acquisition Methodology

1. Camera Data

The webcam provides continuous video frames of the athlete.

The video stream is processed to determine:

- Body landmarks
- Pose
- Body orientation
- Movement trajectory

2. IMU Data

The BN0055 sensors provide motion and orientation information from selected body segments.

The ESP32 collects the sensor readings and transfers them to the Raspberry Pi 5.

3. Processing

The Raspberry Pi 5 performs initial data handling and forwards synchronized information to the main processing and simulation system.

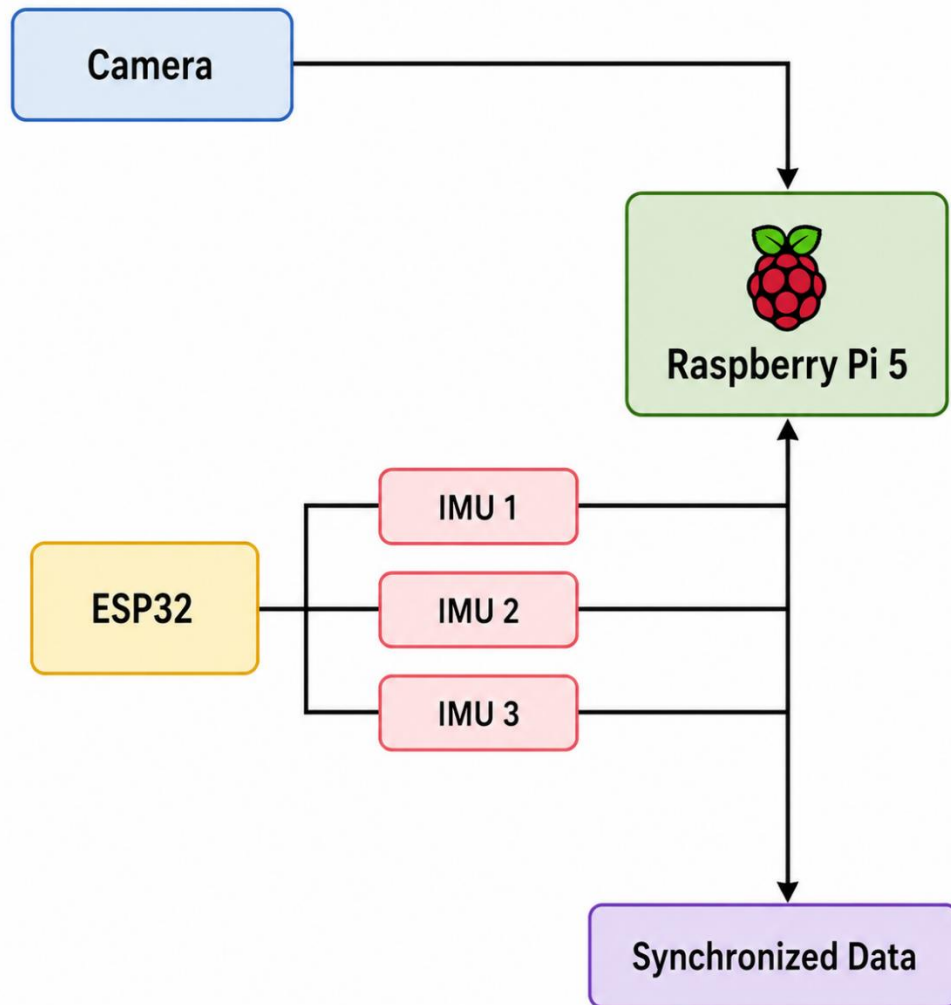


Figure 13: Flow chart for Data Acquisition

V. Application of Proposed Robot in a Societal Context

1. Personalized sports training and performance improvement
2. Affordable performance analysis for emerging athletes
3. AI-assisted coaching and technique optimization
4. Accessible sports-science support for grassroots athletes
5. Performance monitoring and development across multiple sports
6. Supporting the growth of sports and athletic excellence in India

VI. Size of robot:

Since EKAGRATA uses a virtual articulated robotic athlete, the dimensions refer to the proposed virtual robot model used in the simulation environment.

Sr. No.	Parameter	Value (cm)
1.	Length	45
2.	Width	60
3.	Height	175

Table 8: POC Dimension

VII. Timeline for Robot making with milestones:

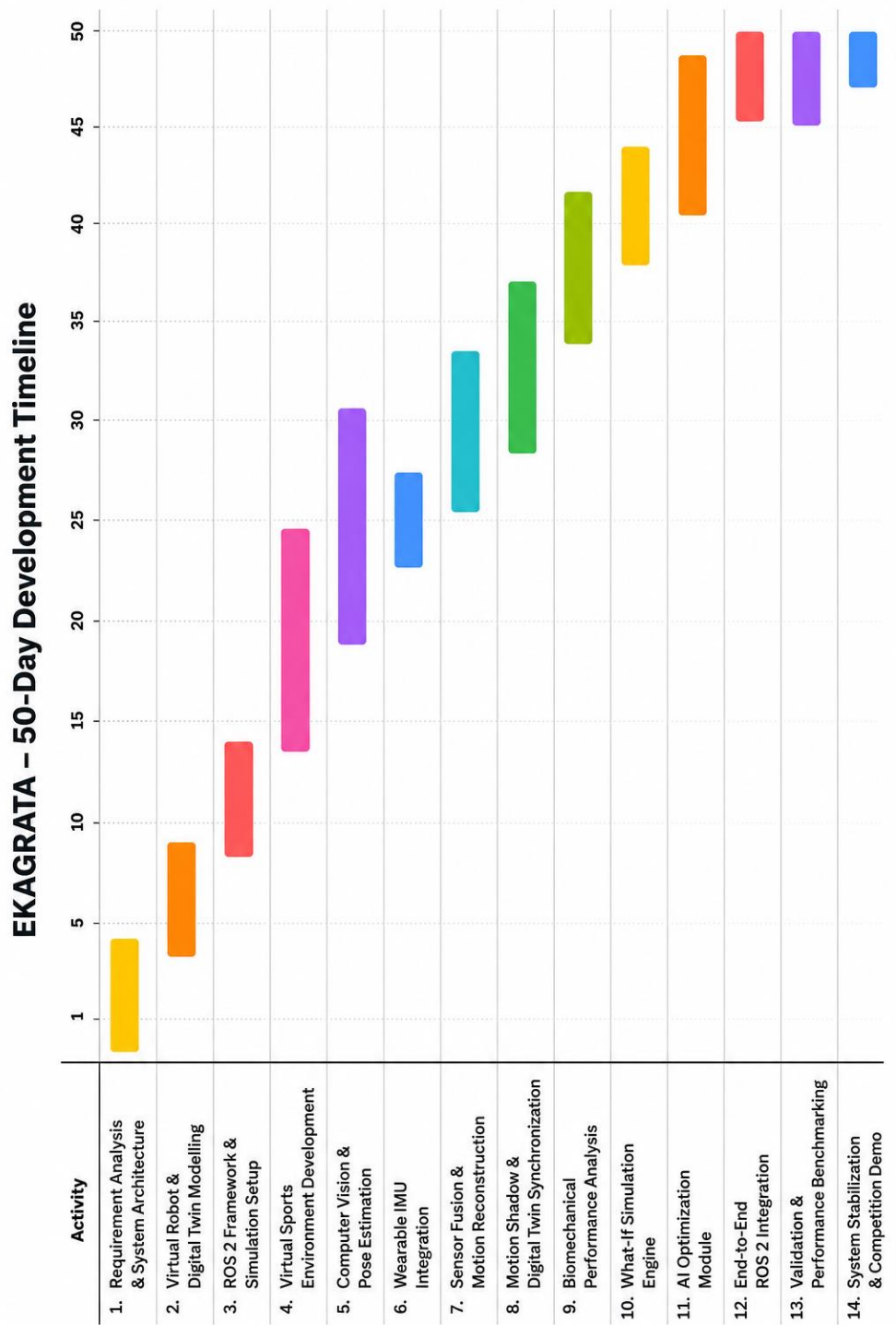


Figure 14:Gantt chart - Divided in activities vs. no. of days

Figure X: Sequence of Activities – EKAGRATA

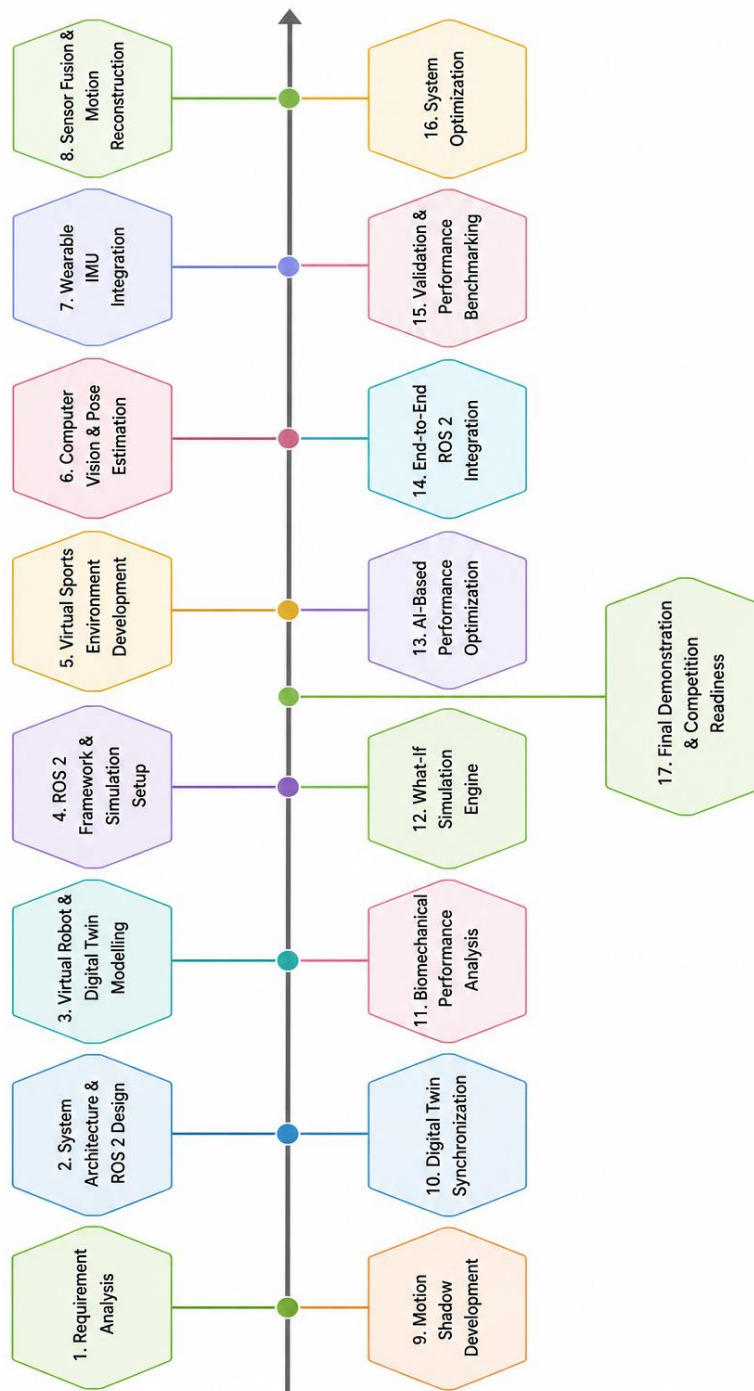


Figure 15: Sequence of activities

VIII. Proposed Outline:

The proposed outline illustrates the complete EKAGRATA system, showing how athlete movement is captured through camera and wearable IMU sensors, reconstructed into a digital twin with a real-time Motion Shadow, evaluated in a virtual sports environment, and compared with optimized motion configurations to generate AI-based performance insights.

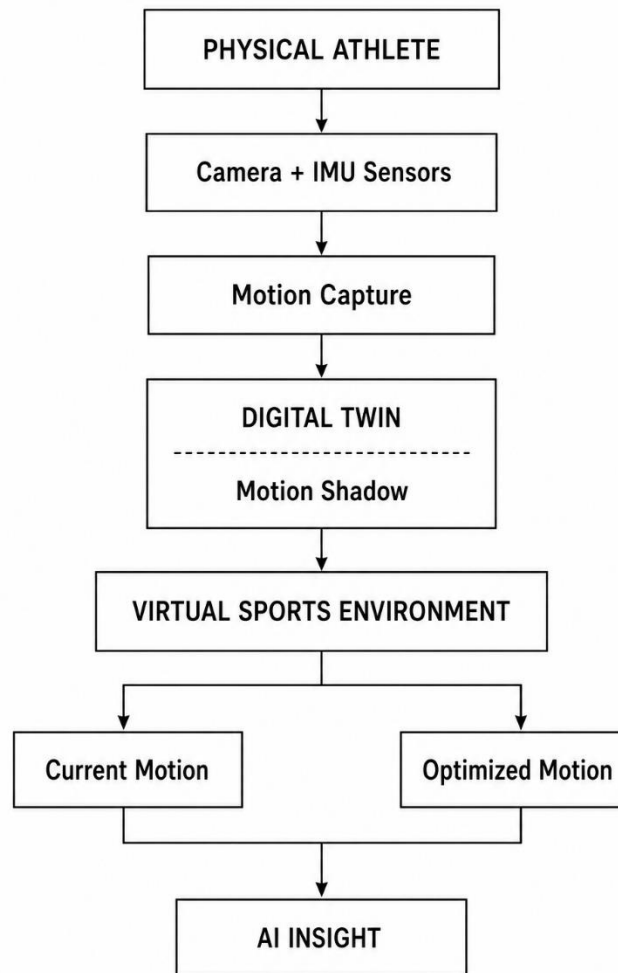


Figure 16: Proposed Outline of EKAGRATA – Integrated Athlete Motion Tracking, Digital Twin and Virtual Performance Optimization

